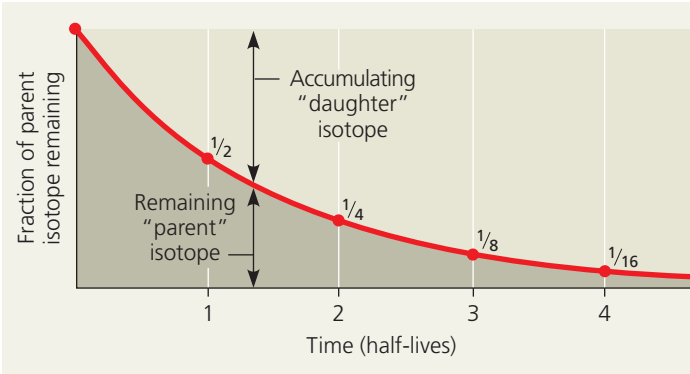


▼ **Figure 25.6 Radiometric dating.** In this diagram, each unit of time represents one half-life of a radioactive isotope.



DRAW IT Relabel the x-axis of this graph in years to illustrate the radioactive decay of uranium-238 (half-life = 4.5 billion years).

isotope to decay (Figure 25.6). Each type of radioactive isotope has a characteristic half-life, which is not affected by temperature, pressure, or other environmental variables. For example, carbon-14 decays relatively quickly; its half-life is 5,730 years. Uranium-238 decays slowly; its half-life is 4.5 billion years.

Fossils contain isotopes of elements that accumulated in the organisms when they were alive. For example, a living organism contains the most common carbon isotope, carbon-12, as well as a radioactive isotope, carbon-14. When the organism dies, it stops accumulating carbon, and the amount of carbon-12 in its tissues does not change over time. However, the carbon-14 that it contains at the time of death slowly decays into another element, nitrogen-14. Thus, by measuring the ratio of carbon-14 to carbon-12 in a fossil, we can determine the fossil’s age. This method works for fossils up to about 75,000 years old; fossils older than that contain too little carbon-14 to be detected with current techniques. Radioactive isotopes with longer half-lives are used to date older fossils.

Determining the age of these older fossils in sedimentary rocks can be challenging. Organisms do not use radioisotopes with long half-lives, such as uranium-238, to build their bones or shells. In addition, sedimentary rocks are often composed of sediments of differing ages. Although we cannot date these older fossils directly, an indirect method can be used to infer the age of fossils that are sandwiched between two layers of volcanic rock. As lava cools into volcanic rock, radioisotopes from the surrounding environment become trapped in the newly formed rock. Some of the trapped radioisotopes have long half-lives, allowing geologists to estimate the ages of ancient volcanic rocks. If two volcanic layers surrounding fossils are found to be 525 million and 535 million years old, for example, then the fossils are roughly 530 million years old.

The study of fossils has helped geologists establish a geologic record: a standard time scale that divides Earth’s history into four eons and further subdivisions (Table 25.1). We’ll describe key events that occurred during these different time periods in Concept 25.3.

Table 25.1 The Geologic Record					
Eon	Era	Period	Epoch	Age (Millions of Years Ago)	
	Cenozoic	Quaternary	Holocene	0.01	
			Pleistocene	2.6	
		Neogene	Pliocene	5.3	
			Miocene	23	
		Paleogene	Oligocene	33.9	
			Eocene	56	
			Paleocene	66	
			Phan-erozoic	Mesozoic	Cretaceous
		Jurassic			201
		Triassic			252
	Paleozoic	Permian	299		
		Carboniferous	359		
		Devonian	419		
		Silurian	444		
		Ordovician	485		
		Cambrian	541		
		Protero-zoic	Neo-protero-zoic	Ediacaran	635
	1,000				
	2,500				
Archaean			4,000		
Hadean			Approx. 4,600		

The Origin of New Groups of Organisms

Some fossils document the origin of new groups of organisms. Such fossils are central to our understanding of evolution; they illustrate how new features arise and how long it takes for such changes to occur. We’ll examine one such case here and in Figure 25.7: the origin of mammals.

Along with amphibians and reptiles, mammals belong to the group of animals called *tetrapods* (from the Greek *tetra*, four, and *pod*, foot), named for having four limbs. Mammals have a number of unique anatomical features that fossilize readily, allowing scientists to trace their origin. For example, the lower jaw is composed of one bone (the dentary) in mammals but several bones in other tetrapods. In addition, the lower and upper jaws in mammals hinge between a different set of bones than in other tetrapods. Mammals also have a unique set of three bones that transmit sound in the middle ear, the hammer, anvil, and stirrup, whereas other tetrapods have only one such bone, the stirrup (see Concept 34.6). Finally, the teeth of mammals are differentiated into incisors (for tearing), canines (for piercing), and the multi-pointed premolars and molars